



AI Data Centers: Big Tech's Impact on Electric Bills, Water, and More

Massive data centers are gobbling up resources across the U.S.—and you may be paying for it



Hyperscale data centers like the QTS Data Center in Fayetteville, Ga. (shown above), are being built across the U.S.

PHOTO ILLUSTRATION: LACEY BROWNE/CONSUMER REPORTS, GETTY IMAGES



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John Steinbach was shocked to receive a \$281 electricity bill in January 2026—a huge spike from the roughly \$100 he'd paid the previous month. “It’s just so far beyond any bill that I’ve ever had,” he says. Steinbach, who has lived in his Manassas, Va., home for nearly 40 years, worries his rates will keep climbing as the outsized electricity demand from AI data centers grows. “They’re building them like it’s ‘Field of Dreams’—build it and the electricity will come—but we don’t see how that’s going to happen.”

The contribution of AI data centers to higher bills is just one of the ways the development boom is affecting consumers. The facilities also compete for critical resources like water and land, and they can lower air quality and increase traffic, often while benefiting from changes to zoning laws and huge tax breaks.

Data centers are not new. Such buildings have been around for decades, housing the servers and other hardware needed to power the internet. But since the introduction of ChatGPT to the public in late 2022, generative artificial intelligence has exploded, requiring mountains of new, power-hungry equipment.

To meet this ballooning demand, and in a race to drive AI's growth, tech giants like Amazon, Google, Meta, and Microsoft have been investing billions to build immense new facilities packed with servers and other equipment. Known as hyperscale data centers, they are far bigger than earlier versions, and are often built in sprawling industrial parks.

Meta alone has dozens of these mega facilities in various stages of development, including its Hyperion campus in rural Louisiana. Originally planned for 2,250 acres, the site recently expanded to 3,650 acres, according to reporting by Fortune—the size of 2,765 football fields, and twice as big as the state's international airport.

3,069 data centers already operate in the U.S., with an additional **1,489** planned or under construction.

Source: Data Center Map, March 2024

Source: Data Center Map, March 2026.

A growing number of them are "hyperscale" data centers. Driven by the AI boom, these sites can sprawl across thousands of acres and consume vast amounts of power. Shown here: **68 hyperscale facilities** that each use at least 50 MW of electricity—as much as a small city. The biggest can use 5 GW or more.

Source: Cleanview, March 2026 (50+ MW demand).

The boom is just getting started. Hundreds more hyperscale facilities may soon be built across the U.S.
Shown: **267 planned** data centers, each using 50 MW or more.

Source: Cleanview, March 2026 (50+ MW demand).

Meta's Hyperion campus in Richland Parish, La., covers **3,650 acres**—twice the size of New Orleans' main airport and four times as big as New York City's Central Park.

Source: Fortune; FAA/NOLA airport; NYC Parks (Central Park).

This buildout is happening all across the country. While the size of the facilities is startling, their energy use is even more dramatic. A January 2026 report by Bloom Energy, an on-site energy company, predicts that U.S. data centers' total combined energy demand—how much electricity they may use when up and running—will nearly double between 2025 and 2028, jumping from 80 to 150

gigawatts (GW). That's like adding a country with the energy needs of Spain in just three years.

The companies leading the AI data center boom say their developments can generate economic growth and jobs, and a number of tech executives recently signed on to a White House-sponsored pledge aimed at limiting their effect on residential electric bills.

However, many local residents and consumer advocates see mainly downsides to the expansion of data centers.

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“People are starting to be really, really aware that these projects tend to be very extractive and bring very little to local communities,” says Kasia Tarczynska, senior research analyst for Good Jobs First, a watchdog group focused on government and corporate accountability in economic development. “And that’s where there’s a huge pushback across the country.”

Consumer Reports is urging the industry to back up its pledges with detailed contracts and progress reports that can be monitored by the public, says Chris Harto, CR’s manager for sustainability advocacy. “This is ultimately about trust. Companies need to show—clearly and verifiably—that they’re paying their own way, not driving up everyone else’s electricity bills or harming the environment.”

As new, and ever-larger, AI data centers continue to spring up, here’s what everyone should understand about their impacts—and a growing backlash.

Electricity Bills Are Rising

It's no secret that electricity costs across the U.S. are skyrocketing.

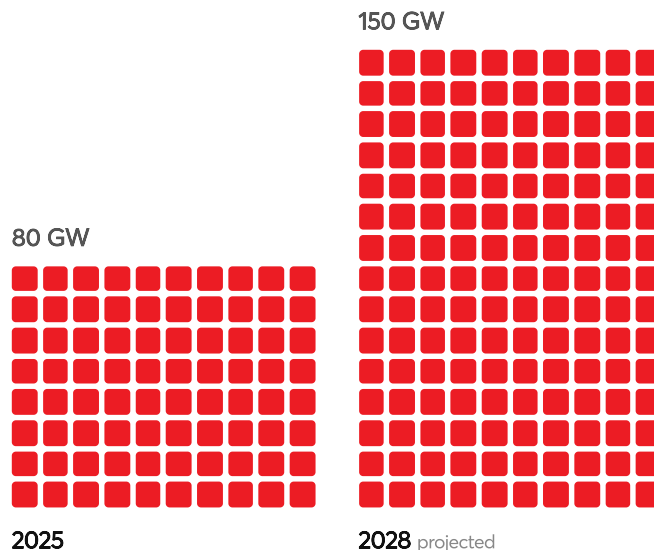
Residential prices jumped 7.1 percent in 2025—more than double the inflation rate—and topped 20 percent in some states, according to federal data. The AI data center rush isn't the only factor driving up prices—others include inflation and an aging power grid—but it's a significant one.

That's because hyperscale data centers demand a nearly unimaginable amount of energy. The International Energy Agency says a typical hyperscale data center might use 100 megawatts (MW), as much electricity as 100,000 households.

And that's at the low end. Meta's Hyperion project in Louisiana, for instance, will need at least 5 GW to run. That's three times as much electricity as the entire city of New Orleans, according to the Institute on Taxation and Economic Policy.

Data Centers Use Vast Amounts of Energy

Power demand, all U.S. data centers



■ = 1 GW = Output of a nuclear reactor =
Power demand of 1 million homes

Source: Bloom Energy/Industry estimates.

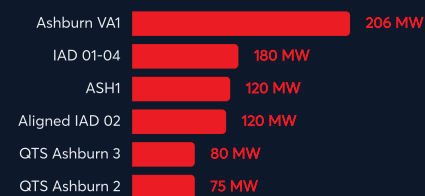
By 2028, data centers could use 12 percent of all the electricity consumed in the U.S., according to the Lawrence Berkeley National Laboratory, and that percentage will grow. More regions could start to look like Virginia, already home to nearly 600 facilities with more than 100 more proposed or under construction. According to a Bloomberg News analysis, data centers accounted for almost 40 percent of the state's total consumption in 2024.

Virginia hosts nearly **600 data centers**, with many more on the way. In 2024, data centers accounted for almost 40 percent of all electricity used in the state.

Source: Data Center Map, March 2026; Bloomberg.

The epicenter is Northern Virginia, where "Data Center Alley" has the world's largest concentration of the facilities. The six most power-hungry facilities use **781 MW of electricity**.

Top Data Center Alley Facilities by Electricity Demand



Source: Cleanview, March 2026.

More data centers can mean higher costs. That same Bloomberg analysis found that areas with high concentrations of data centers saw electricity prices jump 267 percent over the past five years. And John Steinbach isn't the only Virginia resident to see a connection: Nearly three-quarters of voters in the state blame the facilities, largely clustered in Northern Virginia's Data Center Alley,

for rising electricity costs, according to a January 2026 survey conducted by Global Strategy Group and the Chesapeake Climate Action Network Action Fund.

The worry extends beyond Virginia, too. A [November 2025 nationally representative survey \(PDF\)](#) of 2,146 U.S. adults by Consumer Reports found that 78 percent of Americans are somewhat or very concerned that the new data centers being built across the country will make their energy bills go up.

To keep up with the growing demand for power, electric utilities are spending billions to build new transmission lines and power plants. That's one of the main ways new data centers can drive up home electricity bills, says Ari Peskoe, director of Harvard Law School's Electricity Law Initiative and co-author of a March 2025 paper exploring how the public is funding Big Tech's power-intensive facilities. "Utilities are building infrastructure, and then we all pay for it because that's how the utility business model has always worked," he says.

Residential electricity costs are also rising because the rush of new hyperscale data centers wanting to draw power from the grid is spiking demand. That drives up prices for everyone, Peskoe says.

Are Data Centers Driving Up Your Electric Bills?

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Water Demand Is Soaring

Large data centers' rows upon rows of stacked servers and other hardware operate around the clock to power intensive computing tasks like generative AI and cryptocurrency mining—and they run hot. To prevent the machines from overheating, the facilities have historically relied on constantly circulating water to cool their equipment.

In the process, a data center can use up to 5 million gallons of water every day, as much as 16,000-plus average U.S. households, according to Environmental Protection Agency estimates. That's just direct usage for cooling. Generating the electricity to keep data centers powered up requires additional millions of gallons of water, even more than the water used for cooling.



WATER

Data centers around Phoenix use about **385 million gallons** of water per year for cooling—and that doesn't count the water used to generate the electricity they need to operate. The figure will grow as more data centers are built.

Source: Ceres.

WATER

The facilities' direct water consumption may skyrocket to **3.7 billion gallons** annually. That's enough to supply **34,000 homes**.

Source: Ceres, EPA.

Colorado River Reservoir Capacity

Combined Lake Powell and Lake Mead, % of capacity

THE COLORADO RIVER

Hyperscale data centers can worsen water shortages in regions already hit by the 25-year megadrought. Phoenix draws **40 percent** of its water from the Colorado River. The system's two major reservoirs—Lake Powell and Lake Mead—have fallen from roughly **90 percent capacity in 2000** to around **30 percent today**.

U.S. Bureau of Reclamation, RISE Reservoir Conditions. Oct. 1 storage, 2000–2025

0% 2000 2004 2008 2012 2016 2020 2024

What's more, about two-thirds of the data centers built since 2022 are in areas already experiencing water stress, according to a Bloomberg News investigation. That includes parts of Texas. The state is already home to roughly 400 data centers, according to Data Center Map, an industry research company, and it is now contending with the introduction of massive campuses. One planned site, called Project Matador, will house 18 million square feet of data centers—plus four nuclear reactors and a natural gas power plant. The developer, Fermi America, plans to use hybrid air/water cooling systems and says it is investigating the use of recycled water to conserve resources. However, experts at the Houston Advanced Research Center and the University of Houston believe the increased water demand from this project and the dozens of others popping up across Texas will strain supplies in coming years.

In Arizona, another water-stressed state, companies have built, or announced plans to build, 26 data centers since

2022. An analysis by Ceres, a nonprofit advocacy group, found that data centers around Phoenix already use approximately 385 million gallons of water per year for direct cooling needs. And it predicts that amount will skyrocket to 3.7 billion gallons per year once the region's planned data centers come online. That's an 870 percent increase—and it's nearly twice the water needed to supply a city the size of Flagstaff.

Local Impacts Can Be Overwhelming

Hyperscale data center facilities take years to build, and the construction phase can take a toll on community members' daily lives.

In some areas, electric utilities are crossing through people's properties with the new transmission lines needed to supply energy to data centers—with some homeowners even finding massive high-voltage poles going up in the middle of their lawns, just yards from their houses. Homeowners in Georgia have also complained of contaminated well water, damaged roofs, and other problems they blame on nearby data center construction.

They worry about increased traffic, too. Diana Dietz, who lives several miles from a data center campus being developed by QTS in Fayetteville, Ga., says large vehicles are now rumbling through previously quiet areas. "You've got these giant excavation-style dump trucks 2 minutes from my house, nipping by," she says. "You've got those vehicles, you've got huge flatbeds bringing in equipment, you've got huge cement trucks, you've got construction vehicles—all speeding on these residential roads."

Residents face similar problems in other regions, too. Vehicular accidents in the rural area where Meta is building its Hyperion campus jumped by more than 600 percent in the first nine months of 2025 over the previous year, according to an analysis of police records published in November 2025 by Louisiana Illuminator, a local news outlet.

In an emailed statement, a Meta spokesperson says the company is working to "mitigate impacts to the community" by building new roads to divert truck traffic, conducting a traffic study, and collaborating with

contractors and the local police department. The statement also says that the company is “proud to call Richland Parish home” and that “being good neighbors is a top priority.”



A new QTS data center being developed in Fayetteville, Ga., is drawing complaints from nearby residents.

PHOTO: ELIJAH NOUVELAGE/GETTY IMAGES

Once data centers come online, they can also affect local air quality. Because the equipment inside data centers operates day and night, operators often install fossil-fuel generators that can provide power in the event of a shortage or an outage. In late January, with the threat of bitter cold temperatures and impending winter storms in many parts of the U.S., the Department of Energy gave a preemptive green light to data centers and other “major facilities” to run their generators as needed.

Generators powered by diesel and natural gas can be noisy and emit harmful pollutants that have been linked to cancer, heart disease, and respiratory problems. And because many data centers are built close to homes and schools, these emissions can put nearby residents and children at particular risk. According to a recent Kapur Foundation report, 82 percent of California data centers are located in communities already facing poor air quality.

As AI data center construction outpaces the facilities’ ability to connect to the grid in many areas, some are opting to rely entirely on their own generators. Portable methane gas turbines, for example, power Elon Musk’s xAI Colossus 1 and 2 facilities in Memphis, Tenn., affecting communities already overburdened by pollution. (The company did not respond to CR’s requests for comment.)

This “behind the meter” development strategy is increasingly common: Forty-six planned data centers,

with a combined capacity of 56 GW, will avoid connecting to the grid altogether, according to an analysis by Cleanview, an energy data company. Companies that build and operate data centers say that generating power on-site can reduce strain on the grid during peak times, and potentially keep consumer electricity bills in check.

Some of this new power could be supplied by cleaner sources. But for now it will overwhelmingly be generated using fossil fuels.

Data Center Deals Are Often Secret

When a company wants to build a data center in a community, its residents are frequently kept unaware about what's coming until the project is a done deal. Researchers at the University of Mary Washington identified 31 Virginia communities with existing, approved, or proposed data centers and found that the vast majority—25 of the 31—had nondisclosure agreements with local officials. These NDAs often kept the public from learning that a facility was even being proposed, let alone its size or impact on local infrastructure like water and sewage systems.

Data center developers say they need to keep details hidden from other companies while developing their plans. “As is the case with other highly competitive industries, data center companies may use NDAs when considering projects to protect company-specific, competitive information,” Dan Diorio, vice president of state policy for the trade group Data Center Coalition, said in an emailed statement. In fact, he said, “the protections provided by NDAs may result in an increase in the amount and detail of information” shared with local officials about planned water and power usage.

However, consumer advocates say the practice leaves residents in the dark about looming construction projects that will have long-lasting community impacts. “It’s a very mysterious, don’t-look-behind-the-curtain process—and I think that is what is angering people the most,” says Amy Sharma, executive director of the nonprofit Science for Georgia.

Tim Cywinski, communications director for the Sierra Club’s Virginia chapter, agrees: “It’s not a radical thing to

say communities deserve to know if their neighbors are making them sick or driving up their electric bills.”

At the state level, the same lack of transparency extends to economic subsidies for data centers. Many states refuse to disclose how much they are giving out, or to which companies, according to Tarczynska, the senior research analyst from Good Jobs First. When the tax breaks are made public, they are often substantial: Virginia and Texas each provide about \$1 billion per year in data center exemptions, according to the organization. Tarczynska says that data centers are “one of the most subsidized industries we have ever seen” in the group’s 27-year history.

Companies have funded community programs and promised the projects will create jobs, but experts say the trade-offs are rarely worthwhile. While data centers bring short-term construction work, they may provide relatively few permanent jobs once they’re complete. One study, by nonprofit Food & Water Watch, estimated that by the end of 2024, as few as 23,000 people in the U.S. worked in the data center industry. And in November 2025, two business school professors published research finding “no clear evidence that data centers stimulate local growth in tech employment.”

Opposition Is Growing

Federal agencies haven’t done much to regulate the growth of data centers. In fact, an executive order from President Donald Trump last summer streamlined environmental reviews and other regulations in an effort to hasten the buildout.

But individual communities have started vocally pushing back against many projects—with increasing success. Research firm Data Center Watch found that between March and June 2025, community opposition led to \$98 billion in data center projects being blocked or delayed. And a Heatmap Pro review of public records from January 2026 revealed that at least 25 projects were canceled last year in response to local objections.

Demonstrators in Memphis, Tenn., opposed plans by Elon Musk's xAI to use gas turbines to power a data center.

PHOTO: BRANDON DILL/GETTY IMAGES

After months of intense opposition from Indianapolis residents, for instance, an attorney representing Google announced at a September 2025 city-county council meeting that the company was pulling its rezoning proposal for a \$1 billion data center project planned for the southeast side of the city. The room, filled with people holding signs, erupted in cheers. “For a long time, it felt like we were four people with cardboard swords fighting a monster,” one resident told More Perfect Union after the meeting. “But tonight it shows that people power still rings.”

Last fall, political candidates in a number of gubernatorial and other state races pledged to take on data centers and rising electric rates, including in hotspots like Virginia and Georgia. Meanwhile, more than 230 environmental organizations collectively called on members of Congress to place a national moratorium on new data centers until, they wrote, “adequate regulations can be enacted to fully protect our communities, our families, our environment and our health from the runaway damage this industry is already inflicting.”

States such as New York and Maryland have begun to introduce their own legislation that would temporarily halt new data center construction. “It’s time to hit the pause button, give ourselves some breathing room to adopt strong policies on data centers, and avoid getting caught in a bubble that will burst and leave New York utility customers footing a huge bill,” the New York bill’s co-author, Sen. Liz Kruger, said in a statement.

More broadly, in 2026 so far, lawmakers in more than 30 states have introduced over 300 bills on issues related to data centers, including moratoriums, tax incentives, and energy policy, according to Multistate.

Tech companies aren't slowing their planning for new data centers—quite the opposite—but some have begun promising to address consumers' concerns.

In mid-January, Microsoft outlined a broad plan, pledging to cover its own electricity costs, reduce water use, create local jobs, and avoid seeking tax breaks. Anthropic, the company behind Claude AI, made its own promise to cover electricity price increases related to its data center development. And in early March executives from those companies and additional tech firms traveled to the White House to sign a nonbinding "ratepayer protection pledge." Among other measures, companies agreed to "build, bring, or buy" the power needed for their data centers and to pay for upgrades to electric grid infrastructure related to their facilities.

Harvard's Peskoe says the pledge "does nothing to help consumers," since neither the president nor tech companies control who pays for expansions of electrical grids. "Utilities hold the pen on these issues, and utility regulators make the final decisions," he says.



electricity prices for consumers. Anthropic directed us to a tweet by Sarah Heck, the company's head of public policy that read, in part, "In support of the @WhiteHouse rate payer protection pledge, Anthropic has committed to covering 100 percent of electricity price increases that consumers face from our data centers."

But consumer advocates are waiting for the details. "Making a commitment is one thing," says Evan Feeney, CR's associate director of corporate engagement. "Following through is another."

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Nicole Greenfield

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Trending in Data Centers

With AI Data Centers Scooping up RAM, Laptop Prices Could Spike in 2026



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